DEPARTMENT OF APEX INSTITUTE OF TECHNOLOGY

PROJECTPROPOSAL

1. Project Title: -

Liver Disease Prediction Using Clinical Data

2. Project Scope: - (Max 500 words)

The liver is the largest gland in the human body and plays a vital role in metabolism, detoxification, digestion, and immunity. It functions as both an endocrine and an exocrine gland and has a remarkable ability to regenerate. Despite these capabilities, liver-related diseases remain one of the major causes of mortality worldwide, primarily due to late diagnosis and lack of awareness about early symptoms. Many liver diseases progress silently and are often detected only at advanced stages, reducing the chances of successful treatment.

The objective of this project is to develop a predictive model that can identify the likelihood of liver disease using clinical and laboratory data obtained from medical reports. By analysing key liver function parameters such as bilirubin levels, enzyme concentrations, protein levels, and patient demographics, the system aims to assist in early detection of liver-related disorders.

The proposed system will use machine learning techniques to process and analyse clinical datasets. The model will be trained on historical medical data to learn patterns associated with liver disease. Once trained, it will be able to predict whether a patient is at risk based on their test results. This approach can help healthcare professionals make faster and more accurate decisions while also empowering patients to seek timely medical attention.

The scope of this project includes data collection, preprocessing, exploratory data analysis, model selection, training, testing, and performance evaluation. Tools such as Python and machine learning libraries will be used to build the predictive system. The final outcome will be a reliable and efficient model that can support early diagnosis, reduce manual analysis, and improve overall healthcare outcomes.

This project does not aim to replace medical professionals but rather to act as a decision-support system that enhances diagnostic accuracy. Early prediction of liver disease can significantly improve treatment success rates, reduce healthcare costs, and ultimately save lives.

3. Requirements: -

- Hardware Requirements
  1. Laptop/Desktop with minimum Intel i5 processor or equivalent.
  2. Minimum 8 GB RAM.
  3. 10 GB of minimum free space.
  4. Stable internet connection.

- Software Requirements
  1. Google Collab
  2. Overleaf
  3. Python

STUDENTS DETAILS

Name	UID	Signature
Om Singh	23BDA70214	
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APPROVAL AND AUTHORITY TO PROCEED

We approve the project as described above, and authorize the team to proceed.

Name	Title	Signature (With Date)
Vijay Bhardwaj(E12849)	Supervisor	